



Environmental Science 101: Nutrient Runoff and a Watershed — A Full Systems-Analysis Session

Environmental Science 101 · Environmental Science

COURSE-OBJ.ENV5101.2

Explain the mechanism of nutrient-driven eutrophication in a watershed, and evaluate competing interventions using evidence about cost, effectiveness, and tradeoffs. A complete 90-minute session packet with warm-up, guided practice, partner/lab work, independent practice, extension work, assessment, and a full answer key.

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Plan the session

Course and objective

Environmental Science 101 · Environmental Science; COURSE-OBJ.ENVS101.2 — Trace how nutrient runoff moves through a watershed to cause eutrophication, and evaluate competing proposed interventions using cost, effectiveness, and tradeoff evidence.

What is included

Cover; instructor guide; key-term and concept preview; original reading or worked examples; warm-up (2 items); guided practice with organizer (4 items); partner/lab discussion (3 items); independent practice (6 items); small-group extension work (4 items); assessment; complete answer key; access and sources page.

Suggested pacing (90-minute session)

Warm-up 8 min · read both field-report passages + vocabulary 15 min · guided source-mapping organizer 18 min · partner cost-effectiveness discussion 18 min · independent practice 22 min · extension work 5 min · assessment 4 min.

Preparation

Print student pages separately from the answer key, or distribute through a secure LMS. Review the reading/examples and organizer before class. No paid textbook or app is required.

Student directions

Read or work through the material carefully. Support every answer with evidence, work shown, or clear reasoning.



Before you begin

Key terms

watershed — the land area that drains into a shared body of water

eutrophication — nutrient-driven algal growth leading to oxygen loss

hypoxic — low in dissolved oxygen; unable to support most aquatic life

nutrient load — the total amount of nitrogen/phosphorus entering a water body

cost-effectiveness — benefit achieved per dollar spent, not just total cost

runoff — water flowing over land, carrying dissolved and loose material with it

Concept focus

A watershed is the land area that drains into a shared body of water; anything applied to that land (fertilizer, animal waste, sewage) can eventually reach the water through runoff. Excess nitrogen and phosphorus act as nutrients for algae, causing algal blooms. When the algae die, bacteria decomposing them consume dissolved oxygen, creating a hypoxic (low-oxygen) "dead zone" where fish and other aquatic life cannot survive — a process called eutrophication. Because a watershed integrates runoff from many sources across a large area, effective intervention usually requires addressing the sources contributing the most nutrient load, not just the most visible ones, and weighing cost against how much nutrient load each option actually removes.

Model 1 — Tracing the eutrophication mechanism

Fertilizer applied to a field is not fully absorbed by crops; rain carries the excess nitrogen and phosphorus into a stream, which feeds the lake. The added nutrients fuel an algal bloom; when the algae die and decompose, oxygen-consuming bacteria create a low-oxygen zone, killing fish that cannot leave the area in time.

Model 2 — Comparing interventions by cost-effectiveness, not sticker price

Option A removes 2,000 lbs of nitrogen/year for \$50,000 (\$25/lb). Option B removes 500 lbs of nitrogen/year for \$20,000 (\$40/lb). Despite B's lower total cost, A removes more nitrogen per dollar spent — a more cost-effective choice if nitrogen reduction is the goal.



WARM-UP

Warm-up

Answer using what you already know or recall from the previous session.

1. Define "watershed" using the lesson's own wording, and explain why it matters that fertilizer applied far from the lake can still affect the lake.

2. Trace the eutrophication mechanism in order, from excess nutrients to fish kill, using the lesson's own steps.



Field Report, Part 1: Mapping the Sources in Lake Clearview's Watershed

Lake Clearview has experienced increasingly severe summer algal blooms over the past six years, with a major fish kill two summers ago traced to a hypoxic dead zone covering nearly a third of the lake. A team of environmental science students is assigned to map every significant nutrient source draining into the lake's watershed before any intervention can be sensibly proposed.

Their survey identifies three major contributors. First, roughly 40% of the watershed's land is row-crop farmland, where fertilizer is applied each spring; soil testing shows a meaningful fraction runs off into feeder streams after heavy spring rains, before crops have grown enough to absorb it. Second, a cluster of older septic systems serving about 300 homes along the lake's eastern shore was installed decades before current standards, and several are suspected of leaking partially treated wastewater into groundwater that reaches the lake. Third, the town's stormwater system, which was not originally designed with nutrient control in mind, channels runoff from streets and lawns (including fertilized lawns) directly into the lake through a single large outflow pipe.

The students estimate relative nutrient contributions: roughly 55% of the nitrogen load from farmland runoff, 25% from the aging septic systems, and 20% from the stormwater outflow — though they note these estimates carry real uncertainty, since nutrient loading is genuinely difficult to measure precisely across a whole watershed.



Field Report, Part 2: Comparing Three Proposed Interventions

Based on the source mapping, the town council is presented with three intervention options, each with different projected effectiveness and cost, and asked to choose how to allocate a limited environmental budget.

Option 1, a farmland buffer-strip program, would pay participating farmers to leave a strip of unfertilized native vegetation between crop fields and feeder streams, which studies show can capture a substantial share of runoff nutrients before they reach the water. Projected cost: \$180,000 over three years, removing an estimated 8,000 lbs of nitrogen per year once fully established, though it requires voluntary farmer participation and full benefits take two to three years to appear as vegetation establishes.

Option 2, a septic system upgrade program, would provide grants covering most of the cost for the 300 eastern-shore homes to replace aging septic systems with modern, better-treating systems. Projected cost: \$2,400,000 total (homes cover a small remaining share), removing an estimated 2,200 lbs of nitrogen per year, with benefits appearing almost immediately after each system is replaced, though the town cannot force any homeowner to participate.

Option 3, a stormwater retrofit, would add nutrient-capturing infrastructure to the town's single large outflow pipe. Projected cost: \$310,000, removing an estimated 1,600 lbs of nitrogen per year, with benefits appearing as soon as construction finishes, and no dependence on voluntary participation since it is entirely within town control.

The council has enough budget for only one option this cycle, and asks the environmental science team to recommend one, using evidence rather than which problem feels most visible.



1. List the three nutrient sources identified in the Lake Clearview watershed and their estimated share of the nitrogen load.
2. Calculate the cost-effectiveness (dollars per pound of nitrogen removed per year) for each of the three proposed options, showing your work.
3. Based on your cost-effectiveness calculations, which single option removes the most nitrogen per dollar spent?
4. Explain why Option 1's low cost-per-pound alone might not make it the automatically "correct" choice, using at least one specific tradeoff mentioned in the passage.

Partner / lab discussion

Discuss each item with a partner or lab group before writing a joint response.

1. With a partner, discuss: why might the town council be tempted to choose the stormwater retrofit (Option 3) even though it is not the most cost-effective option?

2. With a partner, evaluate Option 2 (septic upgrades): is its high cost-per-pound justified by any benefit the other options don't offer? Explain your reasoning.

3. Explain why the students note their source estimates (55/25/20%) "carry real uncertainty." Why does this matter for the council's decision?



Independent practice

Work independently. Show your work or cite evidence in every answer.

1. Using only the evidence given, write a one-paragraph recommendation to the town council, naming your chosen option and citing at least two specific numbers from the passage.

2. Propose a combined or phased plan using more than one intervention, if the town could secure additional funding beyond this cycle's budget. Justify your sequencing.

3. Explain why "removing more nitrogen per dollar" and "removing more nitrogen in total" are not necessarily the same ranking, using the numbers in this case.



Independent practice

Work independently. Show your work or cite evidence in every answer.

4. Identify one nutrient source NOT mentioned in this case that could plausibly also contribute to a real lake's eutrophication, and explain your reasoning.

5. Explain why "voluntary farmer participation" is identified as a specific risk to Option 1's projected effectiveness, and propose one way the program could reduce that risk.

6. A classmate says "the septic systems must be the real problem, since old infrastructure sounds the most obviously broken." Evaluate this claim using the data in the passage.



Small-group extension work

Use for review or additional application practice.

1. Explain the relationship between a watershed's size/land use and the difficulty of managing nutrient pollution, using evidence from this case.

2. Write a short paragraph (4-5 sentences) explaining eutrophication to a town resident with no science background, without using the word "eutrophication."

Small-group extension work

Use for learners ready for additional challenge or synthesis.

1. Explain why timing (spring fertilizer application before crops are established) makes farmland runoff worse, connecting it to the eutrophication mechanism.

2. Explain what would happen to the lake if the town did nothing and all three sources continued at current estimated rates. Use the term "nutrient load" in your answer.



Assessment

Complete independently to show what you learned today.

1. Self-assessment: which part of this lesson do you feel most confident explaining — the eutrophication mechanism, or the cost-effectiveness comparison — and which do you want to review again?



Assessment · self-assessment

Rate today's work:

4 — I can teach this to someone else.

3 — I can do this independently.

2 — I can do this with help.

1 — I need to revisit this before the next session.



Answer key

1. A watershed is the land area that drains into a shared body of water; because runoff from anywhere in that land area eventually reaches the water, fertilizer applied on farmland far from the lake's shore can still contribute nutrients that reach the lake through streams and groundwater.
2. Excess nitrogen/phosphorus reach the water and fuel an algal bloom; when the algae die, decomposing bacteria consume dissolved oxygen; the resulting hypoxic (low-oxygen) zone cannot support fish and other aquatic life that cannot leave in time.
3. Farmland runoff (~55%), aging septic systems (~25%), and the stormwater outflow (~20%).
4. Option 1: $\$180,000 \div 8,000 \text{ lbs/yr} \approx \$22.50/\text{lb}$. Option 2: $\$2,400,000 \div 2,200 \text{ lbs/yr} \approx \$1,090.91/\text{lb}$. Option 3: $\$310,000 \div 1,600 \text{ lbs/yr} \approx \$193.75/\text{lb}$.
5. Option 1, the farmland buffer-strip program, at approximately \$22.50 per pound of nitrogen removed per year — far more cost-effective than Options 2 or 3.

Accept equivalent, well-supported alternative answers throughout.



Answer key

6. Option 1 depends on voluntary farmer participation (it could underperform if farmers don't sign up) and takes two to three years for full benefits to appear, unlike Option 3's immediate, guaranteed effect once built — cost-effectiveness alone does not capture reliability or timing.

7. The stormwater outflow pipe is a single, visible, directly town-controlled source, which may feel like the most obvious or politically satisfying target to "fix," even though the data shows farmland runoff contributes a larger share of the nutrient load and Option 1 removes more nitrogen per dollar.

8. Answers vary; a reasonable case notes Option 2 may address other concerns beyond nitrogen alone (e.g., pathogen contamination, groundwater quality, property values for the 300 homes), which the pure cost-per-pound-of-nitrogen metric does not capture — a fuller evaluation might weigh these unmeasured benefits against the high cost.

9. Nutrient loading across a whole watershed is genuinely difficult to measure precisely, since it depends on rainfall patterns, soil conditions, and other variable factors; if the true farmland share were significantly different than estimated, the relative case for Option 1 could weaken or strengthen accordingly, so the council should treat the recommendation as evidence-based but not certain.

10. Answers vary; must recommend one specific option and support it with at least two specific figures (cost, removal rate, cost-effectiveness, or timeline) drawn accurately from the passage.

Accept equivalent, well-supported alternative answers throughout.



Answer key

11. Answers vary; must propose a specific combination or order (e.g., start Option 1 for cost-effectiveness while its benefits establish over time, then add Option 3 for its immediate, dependable effect) with a reasoned justification.

12. Option 1 removes both the most total nitrogen (8,000 lbs/yr) and is the most cost-effective per dollar, but in principle these two measures could rank options differently — a very expensive option could still remove more total pounds than a cheap one if its absolute scale were large enough, even while being less efficient per dollar spent.

13. Answers vary; must propose a plausible additional source (e.g., golf course fertilizer, industrial discharge, atmospheric deposition, pet waste) and briefly explain how it could introduce nitrogen or phosphorus to the watershed.

14. If farmers do not enroll enough acreage in the buffer-strip program, the actual nitrogen removed could fall short of the 8,000 lbs/yr projection, since the projection assumes a certain participation level; a way to reduce this risk could include raising payment incentives, simplifying enrollment, or targeting outreach to the farms contributing the most runoff.

15. The data shows septic systems contribute about 25% of the nitrogen load, less than farmland's 55%, and Option 2's cost-effectiveness is far worse than Option 1's; "sounds obviously broken" is an intuitive framing, not evidence, and the passage's numbers point to farmland as the larger and more cost-effective target.

Accept equivalent, well-supported alternative answers throughout.



Answer key

16. Because a watershed integrates runoff from every type of land use within it (farmland, residential septic systems, town infrastructure), managing nutrient pollution requires coordinating across multiple, very different kinds of sources and property owners rather than targeting a single, easily controlled point, making watershed management inherently more complex than a single-source pollution problem.

17. Answers vary; must accurately convey the causal chain (excess nutrients → algae growth → algae death and decomposition → oxygen loss → fish kill) in accessible language.

18. Freshly applied fertilizer sits on bare or newly planted soil with little root structure to absorb it, so heavy spring rain can wash a larger share of it into streams than it would later in the season once crops are established and actively taking up nutrients, directly increasing the nutrient load feeding eutrophication downstream.

19. The total nutrient load reaching the lake would remain high, continuing to fuel algal blooms and the eutrophication cycle; without any reduction in nutrient load, hypoxic dead zones and fish kills like the one two summers ago would likely continue or worsen over time.

20. Answers vary; must name one specific part of the lesson for each half of the question.

Accept equivalent, well-supported alternative answers throughout.



Access, sources, and license

Extension

Ask for a second method, a counterexample, or a short student-led extension of the concept, with a brief explanation of the choice.

Accommodated seating/504

Follow the individual accommodation plan: extended time, breaks, enlarged print, reduced copying, or an alternate response method as documented.

Additional support

Pre-teach key vocabulary or notation before class. Model the first guided item aloud. Accept oral or dictated responses when appropriate.

Multilingual learners

Preview key terms with examples. Allow brief first-language discussion before independent writing. Provide sentence or solution starters.

Course-level alignment · accessed September 18, 2026

COURSE-OBJ.ENVS101.2 — Trace how nutrient runoff moves through a watershed to cause eutrophication, and evaluate competing proposed interventions using cost, effectiveness, and tradeoff evidence. The objective is original wording describing a focused part of the course-level goal.

Course-level learning objective written for this packet; no accreditation, transfer credit, or universal course-equivalency is claimed.

Original reading(s)/worked examples, questions, and answer explanations. No textbook or copyrighted course material reproduced.

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